Project Title

Efficient String Pattern Matching Using the Z-Algorithm: A Comparative Performance Analysis

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project focuses on developing an efficient string matching application using the **Z-Algorithm**, a linear-time pattern matching technique designed for fast searching in large text datasets. The system combines the pattern and text using a special delimiter and preprocesses the resulting string to construct the **Z-array**. The Z-array stores the length of the longest substring starting at each position that matches the prefix of the combined string, enabling the application to identify all occurrences of the given pattern efficiently. The project also implements the **Naïve String Matching Algorithm** to compare its performance with the Z-Algorithm. Performance is evaluated based on execution time and scalability for different text and pattern sizes. The proposed application demonstrates that the Z-Algorithm can significantly improve pattern searching efficiency, particularly for large datasets, by achieving **O(n + m)** time complexity, compared with the **O(nm)** worst-case complexity of the naïve approach. This project highlights the practical benefits of linear-time string matching techniques.

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